WHALETECH TECHNICAL DOCUMENTATION

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Abstract

Abstract

This year, WhaleTech has decided to respond to the Marine Advanced Technology Education (MATE) Foundation's Request for Proposal (RFP) by constructing our newest underwater Remotely Operated Vehicle (ROV), the Whale V. This ROV was designed to accomplish a wide range of tasks in the marine environment, specifically within the Great Lakes. To do this, the Whale V is equipped with an onboard electronics housing, a 3-Function-Manipulator, 3 High Definition (HD) deep water cameras, and 6 BlueRobotics T-200 thrusters. WhaleTech's design will allow its users to complete their mission, whether it be collecting and analyzing the various stages of a jellyfish's life cycle or gaining further information on our marine environment with the Whale V's precision and usability.

Figure 1 Photo of WhaleTech

WHALETECH Robotics
North Paulding High School, 300 N Drive, Dallas, GA 30132 USA
2025 MATE Regional Competition – Ranger Class

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Project Management

Company Profile

WhaleTech is a 5-year-old company located in Dallas, Georgia which specializes in designing and constructing underwater ROVs to help resolve issues outlined by the United Nations Decade of Ocean Science for Sustainable Development (UNDOSSD). The company has 10 members that are split up between the mechanical, electrical, and software engineering departments.

Company Structure

Tyler Pold (CEO), Jacob Banta (CFO), and Ian Osterloh (COO). Along with this base command structure, the company haves implemented leads of the main departments. The lead positions were given out by the most knowledgeable members of each respective department as seen in Figure 2. This structure allows the members with the most experience to be able to teach and guide the newer members of the company according to the skills in the department.

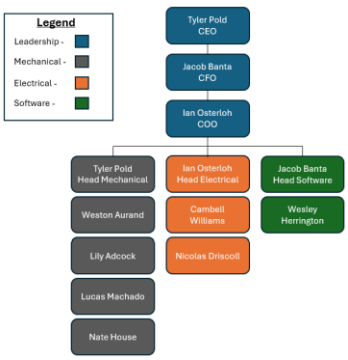

Figure 2 Company Structure

Company Training

This year, WhaleTech consisted of only three returning members and seven newly gained members. Senior members gave the new members a decision on a department they would participate in, which took place at the beginning of our season. Before construction could begin, all new members were taught about safety regulations by coaches and veteran members. After safety precautions were learned and understood, they learned how to effectively use certain tools that are widely used throughout the year such as soldering, angle grinders, and a Dremel.

In October of 2024, WhaleTech began meeting to design and construct the Whale V. Members were split to start discussion on individual components of the ROV. Once the members produced basic designs for their respective components, the leaders of their department would open the floor to them discuss what they built and suggest personal improvements. This allowed us to bring awareness of potential flaws

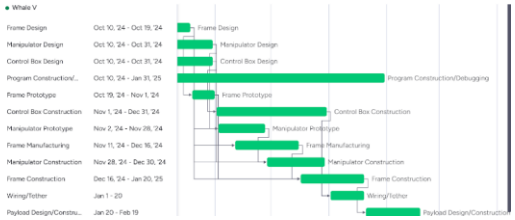

Figure 3 Gnatt Chart

in the design and improve the reliability of the given component. By doing this, the company was able to go through the prototyping phase efficiently and quickly.

Design Rationale

Figure 4 Photo of the Whale V

For the 2024-2025 season, WhaleTech has made their newest product, the Whale V. However, throughout the season, the Whale V has gone through many overall changes. Various aspects of the ROV were taken into consideration, such as its design, software, and electrical systems. With the production divided into sections, WhaleTech was able to focus on certain components before they joined together and finished the design after adjustments.

During the construction of the Whale V's components, WhaleTech utilized the engineering design process. For the prototyping phase of this process, the engineers used everyday materials, such as plywood for the frame, or mock 3d prints. If any problems were apparent within the design during this phase, the company would brainstorm solutions and reciprocate them onto our prototype.

The Whale V's electrical components widely consist of reused parts which were only implemented after testing them through means of test benches such as our ESC and thruster bench, these benches utilized simple wiring and a control code. Some of these reused parts in the control box include the Raspberry Pi 5, voltage regulators, Electronic Speed Controllers (ESC), and the Arduino Uno R3.

Mechanical Rationale

WhaleTech creates ROVs capable of accommodating clientele's needs while remaining adaptable to completing any number of directed tasks. The Whale V's frame is engineered to maneuver through water efficiently with minimal drag. It was precision cut with a CNC (Computer Numerical Control) machine out of features such as neutrally buoyant, high-strength, low-density HDPE (High-Density Polyethylene) which was designed for diligent movement through the water.

The Whale V has three HD underwater cameras placed around the ROV to give the pilots various angles necessary to complete any task in the water. These tasks include

pulling out an anode, identifying and retrieving a medusa stage jellyfish, and removing a damaged thermistor.

Six BlueRobotics T-200 thrusters are mounted on the ROV. One thruster is placed on each corner of the ROV, each being mounted at a 45-degree angle from center to provide omnidirectional movement. In addition, two thrusters are placed on opposite sides of the ROV, producing enough thrust to proficiently navigate the water column.

A 3-function manipulator is mounted underneath the Whale V that has grasping, rotating, and tilting mechanisms to efficiently manipulate objects underwater such as a damaged thermistor or a sacrificial anode. The strategic placement of the manipulator allows for access to objects underneath the ROV and completing its tasks flawlessly.

The components of the Whale IV have been heavily re-evaluated for flaws which are modified in the Whale V. One of the biggest changes made to the Whale V is our small frame design. This decision was made because the company wanted less unused space and to eliminate two thrusters while maintaining how fast it was able to travel.

Testing and Troubleshooting

The process of troubleshooting at WhaleTech begins with isolating the problem. If a specific component does not function as intended, it may be switched out with an identical piece to see if that component is problematic. If the malfunction persists, the wires located at the problem site are tested for continuity, among other testing methods, and then connected components are checked. At the same time, the programming is searched for issues, such as making sure that the correct pins are being used on the Arduino. This method of inspecting the interconnected components continues all the way throughout the ROV and up the tether until the root issue is identified and fixed.

Prototyping is a major part of WhaleTech's ROV manufacturing process and most crucial components, such as the tilt or manipulator, which undergo a series of design techniques. The prototyping process begins with several sketches of the components, for instance, the frame of the robot. Then, a scale model will be created on fusion 360 (see figure 5) and cut it out of cardboard to determine actual shapes and sizes in three dimensions. Once the general shape of the component is determined, a scale model made of plywood is created where components like thrusters can be installed, so WhaleTech can ensure that everything on the ROV properly fits together. With the components, dimensions, and payload positions confirmed, the final product is manufactured.

Mechanical Components

Frame

This year, WhaleTech set out to make a small and compact frame that could be maneuverable in the water with only 6 thrusters to limit the amount of amperage allotted to the thrusters. The decision to make a small and compact frame for our ROV was made because of MATE's RFPs requiring the ROV to get close to a shipwreck while avoiding bumping into it. After the initial design for the frame was discussed and fully 3d modeled (see figure #5), The company assembled a cardboard prototype to envision the size of the frame. Once company members were satisfied with the dimensions, WhaleTech then utilized a CNC cutter to make a plywood cutout to mount components such as the thrusters. This allowed us to make any changes to the dimensions that the ROV may need before the final version was cut out.

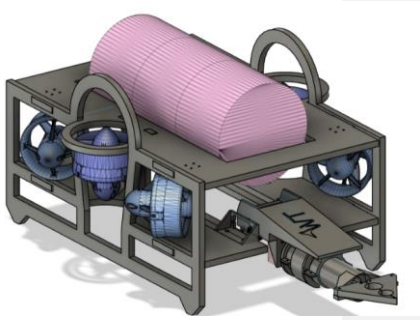

Figure 5 3d Model of the Whale V

The company wanted a frame that was not only small but also able to house all the components that WhaleTech wanted inside of it. Making the frame smaller was difficult due to us wanting 6 T-200 BlueRobotics thrusters with limited obstructions to their water flow, 3 cameras to view all around our robot, a manipulator, and any necessary payloads to complete tasks such as recovering the medusa stage jellyfish. During the cardboard frame phase of prototyping, our engineers adjusted multiple dimensions to ensure plenty of space to be allotted for the components. Further, during the plywood frame phase, the engineers mounted prototypes of the manipulator and the thrusters to ensure that the frame had enough space for our 2 most important components.

WhaleTech decided to use high-density polyethylene (HDPE) for the final frame of the ROV. HDPE was the material used to build the frame because of its proven durability, lightweight, and neutral buoyancy. The final dimensions that were used are 74.93cm x 43.18cm x 28.96cm. These dimensions proved that WhaleTech achieved our goals of being able to support all our components and have a small and maneuverable design.

The Whale V's frame utilizes the mortise and tenon joinery, commonly found in woodworking. This type of joinery ensures the easy repair of any cracked pieces while also

offering a sturdy design. On top of each joint is a piece of HDPE that is screwed into the tenon below to further secure the various parts of the frame together.

Propulsion

Six 3-Phase T200 BlueRobotics brushless motors (see figure 6) propel the Whale V. WhaleTech chose BlueRobotics thrusters for their unmatched reliability compared to other brands and our company's long-term experience with these thrusters. BlueRobotics' T200s have a maximum thrust force of 3.7 kg/f when provided with 12V, as opposed to the T-100s 2.36 kg/f of thrust in the same conditions. Although a higher cost is associated with purchasing T200 thrusters over utilizing old T100s, the T200s produce noticeably higher thrust using the same voltage. They provide the propulsion force necessary for quick and accurate movement across waterways, such as maneuvering to quickly take photos of the shipwreck.

The Whale V achieves omnidirectional movement due to the four horizontal thrusters located at each corner of the ROV. These thrusters are angled at 45 degrees compared to the front and back centerline of the Whale V. This was done to ensure that the front-to-back and side-to-side components of the force produced by the thrusters were equal so that the robot could move in any direction with equal speed. The angles at which the thrusters are mounted are proportional to the sine of the angle and cosine of the angle, respectively, and the only angle with equality between those is 45 degrees. The remaining two thrusters are oriented to thrust vertically on the top plate's sides. They are placed higher on the ROV so as not to interfere with the path of the horizontal thruster's water path.

Figure 6 T200 with shroud

The thrusters on the Whale V have rapid spinning blades that can pose a significant risk to species of marine life that come into contact, or with ocean debris that both pose a risk to the ROV's thrusters. Shrouds were custom-designed and 3D printed specifically for the T200s to form a barrier in which water can pass but foreign objects cannot, therefore limiting the risk of injury or damage. Screws connect the shrouds of the two vertical thrusters through the HDPE surrounding them and were chosen for mounting shrouds rather than epoxy, as used in previous years. In addition, screws allow shrouds to be taken off and used in future builds.

Buoyancy

To keep the ROV balanced, the frame is an HDPE frame which allows it to not only be durable and lightweight, but also neutrally buoyant. Archimedes' principle is used to calculate the buoyancy of the Whale V with its volume. The calculation results in the buoyant force being (number) N. Additionally, WhaleTech decided to make the ROV slightly negatively buoyant.

$$F_b = \rho g V$$

$$F_b = (1000\frac{kg}{m^3})(9.8\frac{m}{s^2})(0.093699425504m^3)$$

$$F_b = 918.25N$$

Total ROV Volume	0.093699425504 m^3
Density of Pool Water	1,000 kg/ m^3
Total Buoyant Force on ROV	918.25 N
Total Buoyant Force on Control Box	53.31 N
ROV Weight	111.034 N

Table 1: Measurements used to determine how much weight needed to be added for the ROV's frame to achieve neutral buoyancy.

Electrical Systems

Onboard Control Unit

The Whale's V onboard control panel is housed within an airtight 6-inch BlueRobotics cast acrylic cylinder. This enclosure is sealed with aluminum end caps, containing fifteen penetrator holes. Inserted inside these ports are BlueRobotics Wet link penetrators, allowing for multiple input and output connections. Furthermore, all ports are sealed off with epoxy to ensure a watertight seal.

Chosen for its interior dimensions, the 6-inch housing provides sufficient space for

maintenance while decreasing the probability of overheating equipment. In addition, reusing this enclosure was a cost-effective decision over purchasing a new one.

Supplied by the surface control box, a 12V power cable is inserted inside the control box and connects to a bus bar. This bus bar outputs 12V to 6 ESCs and the pneumatic pump. Connected to the end terminals of the 12V bus bar is a volt regulator, giving output voltage of 6.5V. 2 servos and a second voltage regulator connect to the 6.5V output. The second voltage regulator steps down from 6.5V to 5V giving power to the Raspberry Pi 5. For power and communications, the Arduino nano connects to a Universal Serial Bus (USB) on the Raspberry Pi 5. A third voltage regulator is used to convert 5V to 3.7V to activate 2 stepper motor controllers.

Figure 7 Tether restraint attached to the Whale V

The Whale V's micro control boards communicate with the programming laptop through a Category 5 enhanced (Cat5e) ethernet cable. The Cat5e ethernet cable is plugged into the Raspberry Pi 5 on the on-board control box. The Pi 5 acts as a hub to relay back data from the cameras, and to send data to the Arduino nano. All signal pins from components on the ROV hook up to the Arduino nano.

In order to keep this system safe from being tugged on, WhaleTech added a custom designed tether restraint CNC cut from HDPE (see figure 6)

Surface Control Box

The Whale V surface control box is used for relaying power and data to the ROV. It's encased in a Monoprice waterproof of ingress protection (IP) 67, measuring 55.88cm x 35.56cm x 20.32cm.

Power connects to the surface control box using Anderson connectors. Attached 30cm away from the power supply, is a 25-amp ATO-type blade fuse in accordance with MATE regulations. Input power is checked by a multimeter, which displays voltage, amperage, and watts. This can help with troubleshooting issues such as voltage drop and over amping. For safety an emergency power switch is installed on a 30-amp relay, which when activated will send 12V through the tether.

Figure 8 ATO Type Blade Fuse

The power switch was made flush with the panel to prevent accidental shut offs and requires a pressurized hold to turn on. Installed on the control box panel is an RJ 45 connector that passes through the laptop ethernet cable to the tether.

To interact with the ROV, two Logitech Extreme 3D Pro joysticks are paired with the

surface laptop. The tether attaches to the control box using waterproof connectors which allow the tether to be fully detachable. Along with this, metal tether restraints are used to properly secure it.

Tether

The Whale V 's surface control box transmits power and data using a 15-meter tether allowing for efficient communication at 1 gigabyte per second (see Figure 9). The cabling is safeguarded in woven polyethylene terephthalate (PET) sleeving, otherwise known as Tech Flex. This abrasion-resistant material offers protection from kinking and exposure to serrated objects to mitigate the risk of inflicting damage upon the cables. Kept afloat by buoyant and flexible polyethylene foam, the tether rests on top of the water as the Whale V makes its advances, keeping away from marine life and machinery, mitigating any risk of entanglement or damage.

Figure 9 The tether of the Whale V

The tether is fixed in place by restraints on both the top and bottom sides, preventing tension in two common areas. A waterproof RJ45 panel mount connector and a four-pin waterproof power connector are installed on the surface side control. These connectors allow the tether to disconnect from the control box. As a result, it is easier to transport and maneuver the two systems independently of each other. Before entering the onboard control box, the tether is guided through two brackets and a latch to act as a tether restraint. This restraint was engineered by WhaleTech employees. It was then CNC cut out of HDPE sheeting and mounted on to the frame (see figure 6). This protects the electrical components from damage by keeping a firm hold of the tether in case of snags. Power to the ROV is supplied through an Automation Direct flexible power cord with four conductive 14AWG wires. Two power and two ground wires were implemented in place of a single power and ground wire to avoid the voltage drop experienced in past designs. In addition, a continuous flexing Cat5e industrial ethernet cable provided by Automation Direct connects the programming laptop to the Raspberry Pi microcomputer within the control box. The main reason this cable was chosen was because it transfers 1000 mbps While also being a better choice over both the Cat5 and Cat6 for having the capacity to transmit data at a faster and more reliable rate than that of the Cat5 which only has a transfer rate of 100mbps, and for WhaleTech's purposes, using a Cat6 cable would be redundant and needlessly more expensive to use where the Cat5e is adequate.

Software Systems

Topside Infrastructure

The Whale V uses a laptop on the surface to manage the onboard system. The laptop reads data from two USB Logitech Extreme 3D Pro joysticks. It then sends data to the onboard Raspberry Pi 5 over Transmission Control Protocol (TCP). The laptop is also used to receive and display the camera feed given to it by the Raspberry Pi 5.

Onboard Software

The onboard system of the Whale V consists of a Raspberry Pi 5, and an Arduino Nano. The Raspberry Pi 5 reads frames from the three cameras and sends these frames to the surface. Once it receives the joystick data from the laptop, it converts the joystick data to microsecond control values for the motors and servos. The Raspberry Pi then sends these values as binary digits over Universal Asynchronous Receiver/Transmitter (UART) to the Arduino Nano.

The Arduino nano in the onboard control box is responsible for delivering microsecond values to the thrusters and servos of the Whale V. The Arduino Nano is more capable of interfacing with components, as it has multiple precise hardware-level timers. In addition to thrusters and servos, the Arduino Nano controls the manipulator's stepper motors and the potential of Hydrogen (pH) sensor. The Arduino Nano receives data through UART and sends data from the pH sensor back to the Raspberry Pi 5 over UART.

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Vision

The Whale V is equipped with three DeepWater Exploration ExploreHD 3.0 cameras, each capable of high definition, low latency, 150° FOV vision. Though expensive, these cameras were chosen for their high performance, and small size. The cameras are capable of high-speed USB 3.0 connections, allowing the Raspberry Pi 5 to utilize the full 1920x1080 pixel resolution at 30 frames per second.

The cameras are placed strategically in three places. The primary camera is mounted on a servo-powered tilt mechanism to allow the pilots to have a wider viewing range, which is helpful for tasks such as identifying the shipwreck. The secondary camera is mounted under the manipulator to get a different angle. This helps the pilots see what the manipulator is doing for precise tasks such as inserting the water sample collection device in the water sample. The third camera is facing downwards to see what's underneath the ROV. This camera is placed specifically for replacing the sacrificial anode.

Figure 10 Front facing camera on a tilt system

Tools

Manipulator

The Whale V is equipped with a 3-function manipulator, including a tilt mechanism, a grasping mechanism, and a rotation mechanism. All these functions work together to complete a wide variety of tasks in the water. These tasks include collecting the polyp stage jellies from the solar panel array and doing so getting information about how we are impacting their life cycle, placing a hydrophone in its designated area to get information about nearby animals around construction sites. A Thermistor is also replaced on a smart buoy and installing a pC02 sensor to gain further information about our marine environment.

This year WhaleTech built a linear actuator based grasping mechanism. This system uses a ⅜ inch threaded rod that gets screwed into a custom 3d printed housing with nuts inserted inside by utilizing a DC motor that can turn at 150 RPM. This motor was chosen over stepper motors because it rotates faster allowing us to grasp objects at a more precise point. The housing then pushes in and out 2 claws that are mounted to a still frame to push them open and closed. The stepper motor design offers multiple advantages over our previous servo based systems by providing a greater grasping force and fixing an issue WhaleTech has faced with gear shearing where the servo's gear would begin to skip, shearing the gear and drastically decreasing the range of motion of the claws.

The manipulator's rotation mechanism utilizes a servo located at the back of the grasping mechanism to rotate the claws. The decision to use a servo instead of a stepper motor or DC motor, which offers multiple advantages, was because of size. Utilizing a servo allows for a smaller manipulator base because a servo length is shorter than that of a stepper motor or DC motor

WhaleTech also wanted to make a more compact tilt mechanism that could move to face directly below our ROV. To do this, a servo was placed on the back of the manipulator with the horn of the servo connecting to an external mount to force the body of the servo to turn with the manipulator. This year's tilt was designed with other servo-based manipulators in mind. This is because these systems worked well and did what they needed them to do but were missing the range of motion that were needed for tasks like replacing the sacrificial anode.

This year's manipulator was designed using fusion 360 (see figure 9) with the 3d printed parts printed out of PLA on our Creality CR-10 Smart Pro. The reason WhaleTech decided to make our own parts instead of having them professionally made is because the company wanted to have flexibility with our design. Another advantage of manufacturing our parts in house is the cost of the part being made is significantly less.

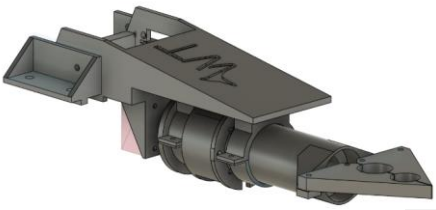


Figure 9 3d model of the manipulator.

eDNA Extraction Device

The eDNA extraction device uses varying components to retrieve the eDNA effectively. Contained within this contraption is a .4mm tubing connected to a 12v peristaltic pump. The pump has a flow rate of 150 ml/min which allows for minimal time spent waiting for the proper amount of eDNA to be pumped. This device extracts the eDNA via tubbing that is connected to a 4mm carbon fiber "straw" with a 2mm inner diameter. To retrieve the sample the manipulator grabs the straw and inserts it into the extraction port. When the pilots activate the pneumatic pump, it creates a suction that collects the sample into a capsule. The eDNA will be pumped into the 3d printed reservoir where the digital pH sensor is located. While it is in the reservoir, WhaleTech can test the pH to ensure that the sample is not diluted. Once an operator has determined that there is enough eDNA extracted they will resurface, where the reservoir can be taken off the ROV and tested further.

Figure 11 eDNA Extraction Device

Jellyfish Recovery Device

WhaleTech was tasked with releasing and catching a medusa stage jellyfish. We came up with a simple design that utilizes a net and a box. On an elevated clip that rests above the camera is a net. This net design captures the jellyfish so when it is brought to the surface, WhaleTech's waterside operator can easily scoop the net into a box and clip the two devices together. This simple device allows the medusa jellyfish to be easily collected and remain submerged in water when on the surface.

Figure 12 Jellyfish Recovery Device

GO-BGC Float

The Remora 2 is an autonomously functioning float that can complete 2 vertical profiles, relay data, and come to full stops for more than 45 seconds. Once the Remora 2 receives a start single, it will descend and complete a halt for 45 seconds then continue its descent. When the Remora 2 reaches the bottom, it will start ascending with a 45 second halt. Once it reaches the surface, it will send a data packet. This data packet contains depth data, pressure data, team name, and team number. The mission station laptop will then use this data to create a depth and pressure graph. This allows WhaleTech to reuse this data to study impacts of global issues like global warming.

The Remora II consists of a buoyancy engine which allows the Remora to complete two vertical profiles with auto stopping functions. The buoyancy engine consists of a pneumatic pump, relay, and bladder. The pneumatic pump and bladder create the buoyancy engine, by using the pump to pull water into the bladder to sink. Flipping the polarity with the relay pushes water out of the bladder allowing it to float.

The control segment allows for wireless communications of depth and pressure data. This section consists of a Pi zero 2W, transceiver, and a BlueRobotics Bar 30 depth sensor. The Pi zero 2W controls all the parts within the Remora II using General Purpose Input/Output (GPIO) pins. The transceiver allows for communication both ways with the mission station. The depth sensor collects depth and pressure data.

The power segment consists of a power switch, battery pack, 5V and 3.3V power rails. The power switch allows for a full power down in case of an emergency. The Battery pack supplies 6V to the pump and the Pi zero 2W. The 5V port off the Pi powers the relay and depth sensor while the 3.3V port off the Pi powers the transceiver.

All sections have custom 3d printed brackets that allow for quick repairs and improve the overall stability of the Remora II, which are housed inside a 3-inch BlueRobotics tube. For waterproofing the tube, 3-inch end caps that have pressure release capabilities.

Fish Recovery Net

As part of our assigned tasks, WhaleTech was required to collect four or more simulated fish species located beneath solar panel arrays. To complete this challenge, our engineers designed a specialized collection box featuring a half-cut front design. This opening allows the fish models to easily slide into a net. To ensure secure and convenient attachment to the ROV, the box includes two support pillars that extend outward and clip directly onto the robot. This makes it simple to attach or remove the collection box during deployment and retrieval

Figure 13 Fish Recovery Net

Safety

Company Safety Philosophy

WhaleTech operates under strict safety policies. Some examples of these policies include senior engineers leading classes, education on the usage of tools, lab protocols, appropriate use of Personal Protection Equipment (PPE), and general work safety per the Job Safety Analysis (JSA). Engineers also work under the supervision of senior engineers to reinforce safety practices. These types of practices have proven well throughout the production of the Whale V.

Training

WhaleTech engineers need to provide the program with help and aid in the process of production in the Whale V. The company also teaches everyone how to handle and use essential tools effectively and safely in case they need to give a hand to others. For example, the company requires all engineers at the beginning of each school semester to take a safety test that must be passed with a perfect score with knowledge in all safety measures such as soldering. The company does this because it benefits all employees, regardless of their specialization.

Vehicle Safety Features

Throughout the production of the Whale V, certain safety features have been necessary to ensure the safety of the engineers. Multiple instances of these features being put in place such as the motors being equipped with custom 3D-designed outer layer of HDPE that surrounds the motor to make sure the motor does not get damaged, there is also “DANGER” signs on the thrusters to make sure that engineers are cautious when around the thrusters, and if an emergency arises while the ROV is in operation, there is an emergency shut-off switch so the ROV can be shut off at a moment's notice, all sharp corners and edges are sanded down to protect personnel. The Whale V follows the mission of working with the ocean in a healthy, symbiotic way.

Lab Safety

The lab is the most important place for WhaleTech Robotics. All employees are expected to return all tools and unused material to their respective labeled containers and dispose of all waste properly. These practices keep the workspaces clear and improve productivity, preventing the need to hunt down tools or materials. To keep the lab safe, all employees follow the company's Job Safety Analysis (JSA), which provides step-by-step procedures to ensure a clean workspace and reasoning for said procedures. Whenever they enter a lab workspace, employees use the JSA to analyze and prevent possible hazards before beginning work to ensure optimal safety while working. WhaleTech Robotics also requires employees to wear appropriate clothing and hairstyles and follow workplace etiquette. All employees must wear closed-toed shoes and form-fitting clothing when in the lab. Further, hair below the ears must be tied up cleanly and moved safely away from moving parts. In addition, any large accessories such as backpacks and large hats must be placed on the ground out of the way or left outside the lab. WhaleTech puts safety first by

prioritizing the protocols inside and out of the lab by having different solutions to different problems.

Poolside Safety

Poolside safety is also needed at WhaleTech to ensure that engineers know what to do during poolside testing. Poolside engineers are required to be able to swim and must be careful near the poolside when operating the ROV, it is also important that engineers ensure power is turned off fully before disconnecting any cables or wires, if not the engineers could get shocked or the robot will fully shut down. Overall, poolside safety is needed to provide the proper instruction of safety during testing or during actual operation over the Whale V.

Finances

Budget

At the beginning of the design phase of the Whale V, WhaleTech prepared an estimated budget of $7,939 based on prior project costs of North Paulding robotics companies. This outline provided the company with a structured schedule to follow regarding company spending so as not to exceed the allotted funding. The projected cost included company expenditures ranging from hardware costs, transportation, lodging expenses, and advertising/technical materials.

A mix of employer costs, fundraisers, and donations supplied WhaleTech with an initial fund of $7,939, of which $5000 were set aside for Whale V expenses, leaving the remaining available for costs such as prop builds, company meals, and other miscellaneous items. Through methods including supporting local businesses through the marketing of consumer goods, WhaleTech raised an additional revenue of $2,313 to support the company's growing expenditures. Sponsor positions were also made available to companies willing to donate to WhaleTech's mission for a returned promise of recognition on company shirts and banners. A total of $600 sponsors' generous donations enabled the company to spend an additional $450 on the production of the Whale V, each given the decision between various degrees of sponsorship.

One of WhaleTech's sponsors, Automation Direct, supported the company through additional means on top of sponsor money, including free and unlimited access to the company's products. This support gave WhaleTech the opportunity to receive free materials such as the Cat5e ethernet cable, saving the company an approximate total of $250 in

spending. The company's decision to reuse specific components after intensive testing was another method for saving money during the Whale V's production, saving an extra $1,000 in the company budget. Components such as the six T200's and three Explore HD deepwater cameras were extensively tested and cross-checked to determine whether buying new products would be an effective use of company funds, both of which were determined to be high functioning enough to be revitalized.

The current project cost of the Whale V, before subtracting items that were donated or reused, is approximately $4,700. After considering all repurposed parts and donated components, WhaleTech spent approximately $3,450. This means the company saved $1,250 in the 2024-2025 production season, approximately $1,550 under the estimated budget, an updated measurement to be used in future WhaleTech productions.

Conclusions

Lessons Learned

WhaleTech undertook every obstacle to develop and improve the company and its product. Throughout the process, each engineer's abilities and soft skills were refined. Skills such as problem-solving, innovation, leadership, and interpersonal communication were taught within the company due to the addition of new members and the repeated confrontation of barriers necessitating improvement. The company's engineers have gained knowledge and skills from years of ROV development, enabling them to create an innovative and reliable product. Problem-solving and innovation are often seen to go hand in hand. Mechanical, electrical and software engineers put the Whale V's components through several stages of testing and troubleshooting to manufacture an efficient ROV. One substantial source of issues.

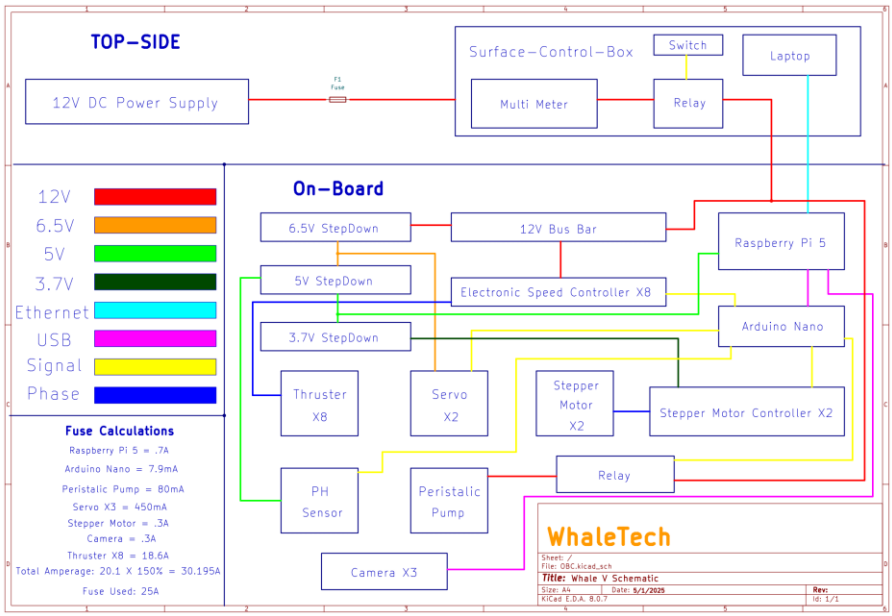
Additionally, WhaleTech's members each developed leadership and teambuilding skills within the workplace, some being promoted into leadership roles based on prior performance and others becoming naturally team oriented. With the company facing organizational issues throughout the production stages of the Whale V, many individuals stepped up or learned to tackle issues on their own or with other company engineers that previously would have required outside input.

Appendixes

Appendix #1 Safety Checklist

PROCEDURE	CHECK
- **Poolside**	
Surface control kill-switch is accessible and functional	
Loose clothing secured, long hair tied back, and close-toed shoes – (Members in appropriate attire)	
Wires are properly managed and other trip hazards removed, cords protected	
Electronics are at a reasonable, large (5ft+) distance from water	
All team members alerted when the Remotely Operated Vehicle or Non-ROV Devices are powered on/off	
- **Workshop**	
Team members using proper personal protective equipment	
Wires are properly managed and other trip hazards removed, cords protected	
Loose clothing secured, long hair tied back, and close-toed shoes – (Members in appropriate attire)	

Appendix #2 Systems Integrated Design (SID)



Appendix #3 Project Costing for 2024-2025

Date	Type	Category	Description	Amount	Running Total
10/1/2024		Beginning Balance	Left over balance from 23/34 fiscal year	$4,264.3	$4,264.38
10/1/2024	Cash Donated	Donation	Company Dues	$3,675.0	$7,939.38
10/1/2024	Re-used	Electronics	Savox waterproof digital servos	($402.00)	$7,537.38
10/1/2024	Re-used	Electronics	T-200 Thrusters, Electronic Speed Controllers	($1,260.0	$6,277.38
10/1/2024	Re-used	Hardware	Continuous Flexing 4-part Cable	($407.40)	$5,869.98
10/1/2024	Re-used	Hardware	Continuous Flexing Cat5e Cable	($99.60)	$5,770.38
10/1/2024	Re-used	Hardware	Weatherproof case, Inline multimeter, Switches, RJ45 ports	($431.16)	$5,339.22
10/1/2024	Re-used	Hardware	6" watertight enclosure	($319.00)	$5,020.22
10/1/2024	Re-used	Electronics	Raspberry Pi 5	($79.99)	$4,940.23
10/1/2024	Re-used	Hardware	DeepWater Exploration ExploreHD 3.0 underwater cameras	($1,140.00	$3,800.23
10/9/2024	Purchased	Equipment	3D printer filament, Worktable, USB-C power supply	($261.82)	$3,538.41
10/29/202	Cash Donated	Sponsorship	Donation from Michael Herrington	$100.00	$3,638.41
10/29/202	Cash Donated	Sponsorship	Donation from Jason & Danielle Herrington	$300.00	$3,938.41
11/11/202	Cash Donated	Fundraiser	Poinsettias Fundraiser	$248.44	$4,186.85
12/5/2024	Cash Donated	Fundraiser	Boston Pork Butt Fundraiser	$684.01	$4,870.86
12/5/2024	Cash Donated	Fundraiser	DQ Fundraiser	$580.00	$5,450.86
12/5/2024	Purchased	Hardware	HDPE plastic sheeting	($232.73)	$5,218.13
1/15/2025	Purchased	Electronics	Peristaltic pump	($86.40)	$5,131.73
1/15/2025	Purchased	Hardware	Raspberry Pi 5 heatsink, Raspberry Pi 2 Zero 2 W	($99.34)	$5,032.39
1/30/2025	Purchased	Electronics	Relays, Wire connectors	($49.10)	$4,983.29
2/3/2025	Purchased	Electronics	Waterproof pH sensor	($76.96)	$4,906.33
2/11/2025	Cash Donated	Donation	Donation from Benjamin Klein	$100.00	$5,006.33
3/10/2025	Cash Donated	Fundraiser	Sheets Fundraiser	$801.00	$5,807.33
3/12/2025	Purchased	Hardware	PVC, PVC sheets, PVC glue, PVC cutters	($617.36)	$5,189.97
3/18/2025	Purchased	Miscellaneous	MATE Regional Registration	($450.00)	$4,739.97

4/21/2025	Purchased	Electronics	Stepper motor, Stepper motor driver	($93.69)	$4,646.28
4/23/2025	Purchased	Equipment	Wire, Wire cutters, Wire connectors, Soldering station, 3D printing filament	($780.21)	$3,866.07
4/26/2025	Cash Donated	Sponsorship	Donation from Magical Vacations Travel	$500.00	$4,366.07

References

BlueRobotics Co. Website (2023, February 24). Retrieved March 2, 2023, from https://bluerobotics.com/

DeepWater Exploration Co. Website (2023). Deepwater exploration. Retrieved March 2, 2023, from https://dwe.ai/products/explorehd

Acknowledgments

As WhaleTech prepared for the 2025 competition, individuals and businesses provided support to the company by donating money, time, and expertise. WhaleTech Robotics would like to thank the following for their continuous help and support.

- MATE Center and Gray's Reef National Sanctuary for creating the 2024 missions and organizing the competitions.
- MATE sponsors (Marine Technology Society, Marine Technology Society ROV Committee, National Oceanic and Atmospheric Administration (NOAA), National Science Foundation (NSF), Oceaneering, Marine Technology Society, Teledyne Marine, Martin Klein, Eastman Foundation, BlueRobotics, Bulgin, Kingsport,

Eastman foundation, Global Ocean Biogeochemistry Array, Niswonger Foundation, Fritz Stahr and Erin Moore, Blue Trail Engineering, and Motorola Solutions Foundation.

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- Thank you to WhaleTech's sponsors for their generous support during the 2023-2024 build year.
- Geoff Gardener and Brandi Pold, thank you for providing their time and team management skills.
- Member families for their encouragement and support throughout the year.
- Mentors – Geoff Gardener, Jason Herrington, and Rocco Leach for investing your time and energy to work on increasing our knowledge.